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Short Communication

Voluntary walking related joint movement training with targeted epidural electrical stimulation enabled neural recovery for individuals with spinal cord injury

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Spinal cord injury (SCI) poses a substantial medical and social challenge, frequently leading to partial or complete loss of motor function below the site of injury [1]. The disruption of descending neural pathways and the consequent loss of neural connectivity impair voluntary movement and interfere with sensory and autonomic functions. Traditional therapeutic strategies have primarily centered on compensatory methods, helping individuals with SCI adapt to their impairments [2]. However, recent advances in neurorehabilitation are shifting the paradigm toward restorative treatments that seek to re-establish neural connections and promote functional recovery.

Epidural electrical stimulation (EES) has emerged as a highly promising intervention for SCI [3,4], involving the application of electrical currents to target the residual neural circuits that remain intact post-injury. A fundamental mechanism underlying the potential of EES in functional recovery is grounded in neuroplasticity [5,6], the ability of the nervous system to reorganize its structure, function, and connections in response to intrinsic or extrinsic stimuli. Targeted electrical stimulation facilitates synaptic plasticity and promotes the strengthening of existing neural pathways while supporting the formation of new circuits and thereby enhancing motor recovery [1].

The therapeutic potential of EES is supported by a growing body of evidence. Human case studies and clinical trials have demonstrated significant improvements in motor function restoration, even in individuals with chronic SCI [3,4]. However, the challenge of restoring complex motor tasks, such as walking, requires precise control over individual joint muscles, which has not been fully explored.

The diverse symptoms following spinal cord injury result in individualized joint movement deficiencies in individuals with SCI, which can potentially lead to complete failure in walking rehabilitation outcomes. Additionally, the complexity of physiological and anatomical structures prevents epidural stimulation from achieving full activation accuracy of walking-related muscle synergies. Therefore, we propose that training combined with targeted neuromodulation and residual volitional control is essential for addressing individual joint movement deficiencies, as it may harness neuroplasticity to restore walking function.

All experiments were conducted as part of the ongoing clinical feasibility study "Closed-loop functional spinal cord stimulation in patients with spinal cord injury (COFUN)", which aims to evaluate the effectiveness of EES in facilitating ambulatory rehabilitation post-SCI. This study was registered with ClinicalTrials.gov (NCT04969042) and was approved by the Beijing Tsinghua Changgung Hospital Ethics Committee (No. 21161-2-02). Two individuals with SCI, one classified as AIS B and the other as AIS C (Table S1

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online), were enrolled in the study, and both signed written informed consent prior to their participation. Before enrollment, several neurological evaluations were conducted, and a dedicated rehabilitation physician accompanied our participants during daily training to provide support and monitor safety issues.

Pre-operative magnetic resonance imaging was performed to determine the optimal position for lead paddle implementation. Participants underwent implantation of a 32-electrode paddle array (Beijing PINS Medical Co., Ltd., model L3255) in the epidural space of the spinal cord, along with two implanted pulse generators (IPGs) of investigational model (Beijing PINS Medical Co., Ltd., model G122RS). The entire surgical procedure was conducted under general anesthesia, with intraoperative electrophysiological signals monitoring to ensure optimal paddle array positioning, specifically targeting muscles relevant to motor function restoration.

During the rehabilitative phase, electromyography (EMG) signals were recorded using Noraxon Ultium System at a sampling rate of 2000 Hz. All EMG data was band passed with a third-order Butterworth filter with cutoff frequency between 25 and 250 Hz to ensure signal fidelity for subsequent analysis. Kinematic information was recorded using Noitom Perception Neuron motion capture system with a sampling rate of 100 Hz.

The primary objective of this study was to restore walking function in participants with SCI through our innovative rehabilitation program. In addition to conventional rehabilitative trajectory progressing from standing to ambulatory exercises, a tailored approach emphasizing the functional recovery of walking related joint movement (WRJM) deficiency has been conducted to achieve this. Therefore, the training protocol incorporated self-training of WRJM exercises parallelly with walking restoration parameter mapping and walking rehabilitation (Fig. 1a). Notably, this approach made full use of the participants' autonomous practice time for activities related to walking function restoration.

The self-training protocol began with the identification of optimal stimulation parameters (Fig. S1 online) to concurrently activate key muscle groups for WRJM by exploring the spinal cord innervation relationships. Initial stimulation intensities were recorded, and a safety margin was established by setting an upper limit of 15 % above the baseline amplitude (Fig. 1b). No lower limit for stimulation intensity was set to allow our participants to voluntarily reduce the intensity based on their motor function status. Movement performance, total and extremity fatigue, participation level and lower limb spasticity scale were evaluated daily (Table S2 online). The configuration of stimulation parameters and training amount for the following day may be adjusted based on the evaluation and participants' physical/mental condition. Following these guidelines, participants were instructed to perform 100–150 repetitions of WRJM independently (P1: hip flexion, P2: knee extension and ankle dorsiflexion) for each leg daily, 4–5 days per week (Fig. 1b and Video S1 online), with the autonomy to adjust the stimulation amplitude within the prescribed safety parameters. Post-training muscle condition evaluation and relaxation using deep muscle stimulator were conducted on related muscles namely iliopsoas, rectus femoris and tibialis anterior to ensure that participants' muscle condition could maintain sustainable training.

In terms of functional outcomes, both participants demonstrated neural recovery, as evidenced by the activation of targeted joint movement muscles without EES. Over the five-week training program, the first participant exhibited the ability to autonomously reduce stimulation amplitude by an average of 50% in hip flexion parameters bilaterally (Fig. 1c). This indicates progressive physical and neurological improvements in the iliopsoas muscle and substantiates the efficacy of tailored involvement in the rehabilitative process. Electromyographic spectrogram analysis confirmed a comprehensive frequency distribution comparable to that of healthy individuals (Fig. 1d). Additionally, participant P2, who was initially unable to induce any left leg muscle activation, achieved knee extension and ankle dorsiflexion muscle activation without EES after training (Fig. 1e and Fig. 1f). Concurrently, assessments of motor control ability without EES were conducted by periodic EMG recording. Participant P1 exhibited improvement in iliopsoas muscle activation compared to baseline levels recorded at the commencement of the recovery process (Fig. 1g). Participant P2's chronological EMG recording showed that the rectus femoris muscle responsible for left knee extension regained its ability after approximately 2 months training. The subsequent training period allowed her to achieve knee extension control with more precision, evidenced by increased activity in knee extension relevant muscles (rectus femoris) and decreased activity in irrelevant muscles (tibialis anterior and soleus) (Fig. 1h).

Given that the primary goal of WRJM training is to achieve outcomes of walking restoration, we conducted additional investigation into effectiveness of self-training to attain this objective. In participant P1, this individualized training elicited spindle-shaped EMG signal morphologies in both limbs, similar to those observed in ambulatory healthy subjects. EMG signals recorded during walking (by applying parameters illustrated in Fig. S2 online) from the iliopsoas, rectus femoris and biceps femoris muscles exhibited spindle-shaped morphologies, analogous to those observed during self-training sessions (Fig. 2a).

Additionally, the joint kinematic changes observed in participant P2 during walking restoration further confirmed the effectiveness of WRJM self-training. Initially, participant P2 was completely unable to achieve left knee extension autonomously, posing a significant challenge for walking restoration as knee extension is essential for providing weight-bearing capacity during the stance phase of gait. After two months of focused training, the participant's rectus femoris muscle activation increased significantly and successfully regained the ability to perform this specific motion, which helped to achieve gait stability and weight-bearing ability (Fig. 2b right and Video S1 online). Interestingly, we tested various other parameters for facilitating stance during walking restoration, but none showed superior performance compared to those used in self-training (Fig. S3 online).

Furthermore, our protocol optimized gait performance during the walking restoration process. Foot clearance during the swing phase of gait is crucial for efficient forward progression and preventing trips and falls. Therefore, participant P2 underwent ankle dorsiflexion self-training with EES. After one week of training with the parameters shown in Fig. 2c left, participant P2 achieved significant increase in tibialis anterior muscle activation and improved her foot clearance performance during the swing phase, resulting in a more natural gait pattern (Fig. 2c right and Video S1 online). This result further validated the capacity of our training approach to facilitate tailored improvements in walking restoration.

By leveraging the principles of neuroplasticity through precise electrical neuromodulation and targeted rehabilitative tasks, there is significant potential to restore walking function in our participants. The concept of self-training WRJM with targeted EES represents a pivotal advancement of therapeutic strategies. Concentrating on individual joints enables refined movement control and addresses specific functional deficits caused by complex nature of spinal cord injuries. Furthermore, this specificity is believed to be critical for inducing localized neuroplastic alterations and for providing a clear sensory-motor feedback loop, which is essential for motor learning within the central nervous system [6]. This self-training methodology empowers individuals with SCI to take an active role in their rehabilitation, ensuring that exercises are both repetitive, which is essential for neuroplasticity, and intrinsically motivating and adaptable to individual progress and preferences [1]. This individual-centered approach aligns with

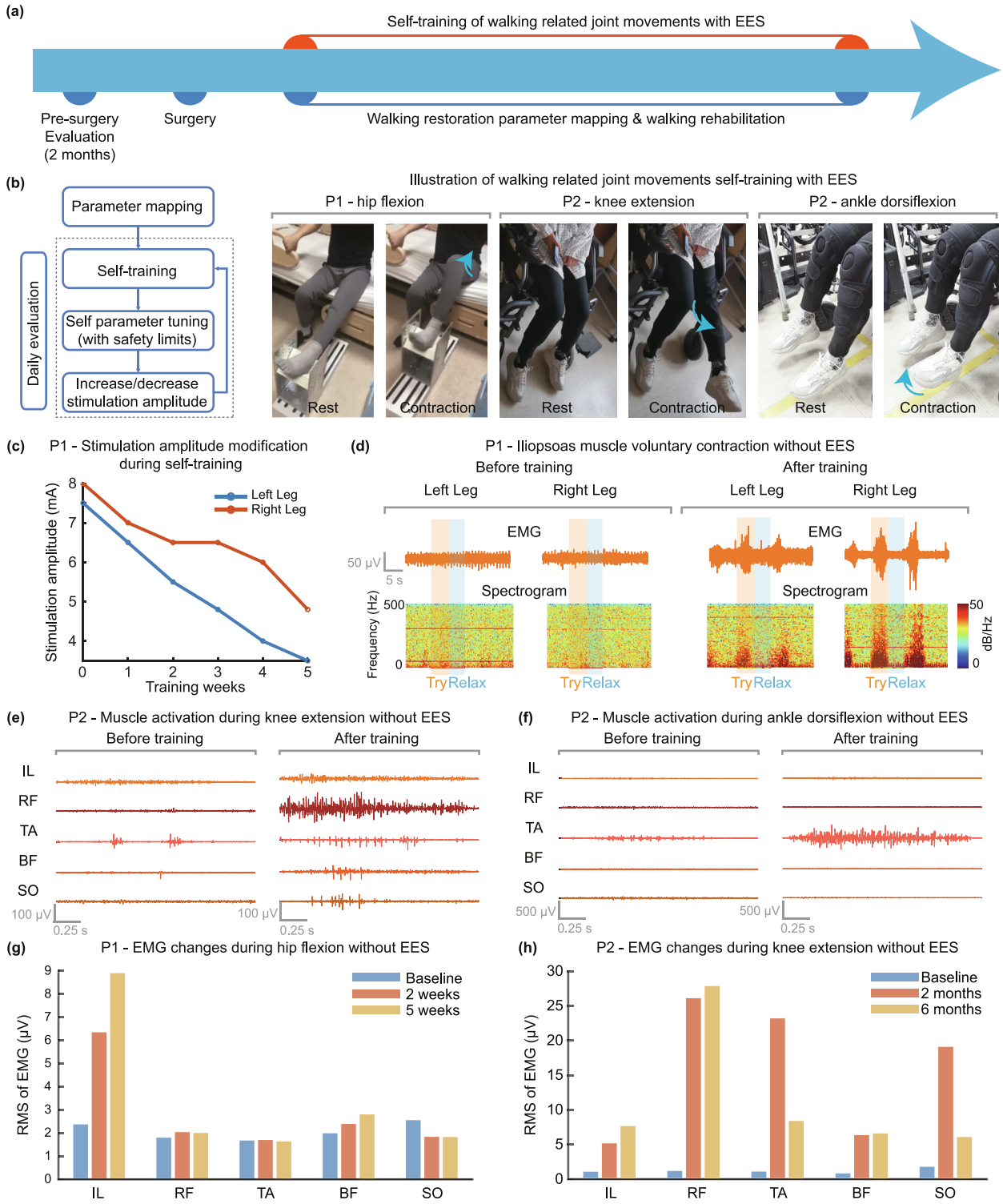


Fig. 1. Study design and functional outcomes following walking related joint movements self-training with EES. (a) Rehabilitation timeline combining walking related joint movement self-training program. (b) Walking related joint movement self-training program pipeline and illustration of self-training process. Safety limits are pre-settled with therapist focused on stimulation amplitude tuning. Daily evaluation is in parallel with self-training to guide training plan adjustment. (c) Stimulation amplitude voluntarily reduced by participant during self-training (P1). (d) Electromyographic and spectrographic performance of iliopsoas muscle voluntary contraction without EES before and after self-training process (P1). (e) Electromyographic changes before and after left knee extension self-training without EES (P2). (f) Electromyographic changes before and after left ankle dorsiflexion self-training without EES (P2). (g) EMG chronological changes during hip flexion training without EES (P1). (h) EMG chronological changes during knee extension training without EES (P2). BF: biceps femoris; IL: iliopsoas; RF: rectus femoris; RMS: root mean square value; SO: soleus; TA: tibialis anterior.

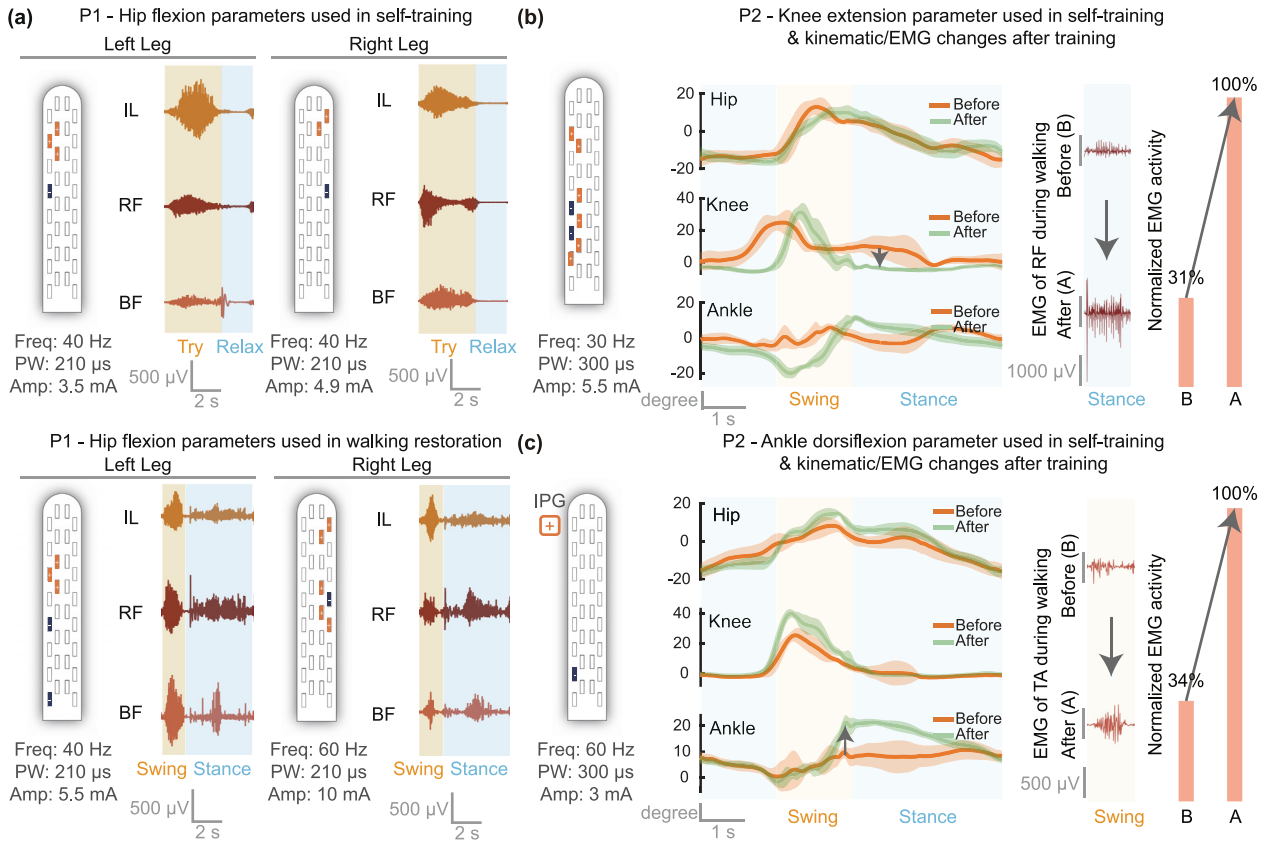


Fig. 2. Kinematic and electromyographic changes during walking restoration after walking related joint movements self-training. (a) Self-training of hip flexion resulted in natural EMG morphology during walking restoration (P1). (b) Kinematic and electromyographic changes in the knee joint and rectus femoris muscle after knee extension self-training (P2). (c) Kinematic and electromyographic changes in the ankle joint and tibialis anterior muscle after ankle dorsiflexion self-training (P2). Amp: stimulation amplitude; BF: biceps femoris muscle; Freq: stimulation frequency; IL: iliopsoas; PW: stimulation pulse width; RF: rectus femoris; TA: tibialis anterior.

contemporary rehabilitation principles that emphasize personalized care and active engagement.

In this study, we propose a strategy to enhance functional recovery in individuals with SCI by identifying a pattern of motor regulation that synergizes with supraspinal control signals [5,6]. Previous studies have also underscored the significance of volitional locomotion and voluntary movement [7,8], emphasizing the necessity of task-specific training for functional recovery based on volitional control [9–11].

Notably, by allowing participants to autonomously adjust their stimulation amplitude, participant P1 significantly reduced his stimulation amplitude (Fig. 1c), while participant P2 achieved muscle activation without EES. We hypothesize that these outcomes are associated with both strengthened muscles and enhanced synergies between EES and volitional control. Although spinal circuits below the lesion are often intact after SCI, they lack sufficient excitation and supraspinal control to achieve functionality [1]. EES recruits proprioceptive afferent fibers and increases the excitability of spinal circuits [3]. When combined with intensive task-specific training [10], the responsiveness of related spinal circuits may be enhanced, facilitating residual commands from the brain to elicit muscle activities [12]. Thus, this paired corticalspinal stimulation may induce task-specific plasticity, enhancing residual neural connections, reducing the dependency on EES for motor neuron activation, and promoting spinal cord automaticity in individuals with SCI [9]. During parallel training with walking restoration, the induced neural recovery was further enhanced by sensory inputs, such as weight-bearing information, leading to achievements in overall walking restoration outcomes as observed in our study.

Moreover, various conditions associated with SCI may influence the outcomes of neural recovery using the method presented in this study [13]. In comparison to participant P2, although participant P1 achieved iliopsoas activation after training, actual joint movement without EES was not achieved. This discrepancy may be related to the etiology, level, and severity of the injury. As detailed in Table S1 (online), participant P1 exhibited lower sensory and motor scores at baseline evaluation compared to participant P2 (110 vs. 135 for sensory scores and 0 vs. 7 for motor scores). Additionally, we observed that for participant P2, WRJM training with EES produced superior effects on the left leg relative to the right leg. This finding corresponds with the lower extremity motor scores (left 6 vs. right 1). These findings suggest that residual neural functional connections may influence the outcomes of our methods.

Regarding muscle strain and fatigue monitoring, although palpation was employed to evaluate participants' muscle states after training, this study acknowledges a limitation in quantitative evaluation. Consequently, future protocols should incorporate additional diagnostic methods, such as electrical impedance myography evaluation [14], to improve daily muscle monitoring practices. This proactive approach seeks to safeguard musculoskeletal status and mitigate potential risks associated with the rehabilitative protocol.

Concerning walking restoration, it is important to note that even after participant P1's hip flexion training, the stimulation parameters still need to be fine-tuned for effective application to ambulatory function restoration (Fig. 2a). This phenomenon is attributed to the interference of the spatial–temporal electrical field [15]. EES generates an electrical field that targets the dorsal

root area to activate lower limb muscles. However, these electrical fields may interfere with one another. For instance, simultaneous stimulation for weight support in the left leg and swing in the right leg could induce such interference. Therefore, additional anodes may need to be added around the applied cathodes to mitigate this observed interference.

Finally, while integrating EES with targeted rehabilitation tasks shows promise, it is not devoid of complexities. The optimization of stimulation parameters is crucial for attaining desired motor outcomes. Additionally, selecting appropriate candidates for EES therapy and tailoring interventions to individual injury profiles require careful consideration. These aspects underscore a necessity for an interdisciplinary approach that combines medical expertise, engineering techniques, and therapeutic insight to address the complexities inherent in optimizing EES applications for spinal cord injuries.

Conflict of interest

The authors declare that they have no conflict of interest.

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Author contributions

Xi Zhang and Luming Li contributed to the conceptualization of the manuscript. Xi Zhang, Yang Lu, Boyang Zhang, Bozhi Ma, Yu Pan, and Guihuai Wang were responsible for establishing the clinical platform. Xi Zhang, Yang Lu, Boyang Zhang, Hongda Li, Qingyu Yao, Junhong Lim, Yunyue Wei, Kaibo He, Chenhui Zuo, Yanan Sui, and Mingshan Ran contributed to the execution of the experiments. Xi Zhang was primarily responsible for writing the manuscript, Yang Lu, Boyang Zhang, and Luming Li edited the manuscript, and all authors approved the final manuscript.

Appendix A. Supplementary materials

Supplementary materials to this short communication can be found online at https://doi.org/10.1016/j.scib.2024.09.036.

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